

Life Cycle Assessment of Exact 31-mer Counting on an HPC Node: Carbon Payback of an AI-Assisted Optimisation

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Minimum of 4 double-column pages, excluding guidance text, references and appendices.

Compiled September 29, 2026.

Abstract—

I. INTRODUCTION

II. RELATED WORK

III. LIFE CYCLE ASSESSMENT

A. Goal and scope definition

The goal and scope definition fixes what is studied, the functional unit every later figure refers to, the boundary of the system and the method that turns inventory flows into an impact [1], [2].

Goal. This study quantifies the carbon footprint of one execution of a baseline 31-mer counting job. That footprint is the reference against which the AI-assisted optimisation of Section IV and its carbon payback in Section V are measured.

The job. The code is `kc-pyl.py` from the public repository `lh3/kmer-cnt` [3], frozen at commit `e2574719c9cfb784915d80eb5828e78dfae4cfd7b`. It is a single-threaded Python 3 program. It reads FASTA from standard input and counts every canonical 31-mer, the lexicographically smaller of a 31-mer and its reverse complement. It skips any 31-mer that contains N. It prints a histogram with 255 lines, one per count from 1 to 255, and pools every count above 255 into the last line. The input is `M_abscessus_HiSeq_10M.fa.gz` from the repository's v0.1 release [3]: 187 693 955 bytes, SHA-256 `dd2ef49aebabfed1574b55fd002025f29b2fda88c70b2095f5e1b3a822cee736`. It is decompressed on the fly by `gzip`. The program upper-cases only the last record of its input. No sequence line of this input holds a lower-case base (checked on 28 September 2026), so this does not affect the result. The job is submitted with the Slurm script `run-baseline.sh` of the project repository at commit `253d93ec726b82670c8f4b84da20e9528c9a59ef`. Every run writes the same 1 972-byte histogram, SHA-256 `06a32de77200121a397050e015a257953dc956b7854027a1ed73b15afc12bf42`.

The machine. The job runs on the Aion cluster of the University of Luxembourg HPC platform, partition `batch`, QoS `normal`, on the regular compute node

`aion-0127`. An Aion node holds two AMD EPYC 7H12 processors of 64 cores each, 256 GB of DDR4 memory, a 480 GB local SSD and one HDR100 InfiniBand port [4]. The loaded modules are `env/release/default` and `lang/Python/3.11.5-GCCcore-13.2.0` (Python 3.11.5), as recorded in every run's log.

The allocation. Each run holds one node exclusively: one task, 128 cores, all the memory Slurm can reserve on the node (`--mem=0`, recorded as 224 000 MiB) and a walltime limit of 30 minutes. The program uses one of those cores. The node is reserved exclusively so that no other job shares it. As a consequence, the whole node, not one core, is attributed to the job.

The user. The job is submitted with `sbatch` from an Aion login node by the author, through a standard student account.

How many runs. The functional unit is one run. Five runs are measured (Section III-B). The job is assumed to be executed 100 times per year, an assumption listed in Section VI.

Functional unit. The GWP100 carbon footprint of one execution of `kc-pyl.py` at commit `e2574719` of `lh3/kmer-cnt`, counting exact canonical 31-mers in `M_abscessus_HiSeq_10M.fa.gz` (187 693 955 bytes, SHA-256 `dd2ef49a...e736`), submitted by a student user on Aion, partition `batch`, holding one exclusive regular node (2 × AMD EPYC 7H12, 128 cores, 224 000 MiB reservable memory) for a median walltime of 368 s.

Method. One criterion: a carbon footprint in CO₂e. The characterisation method is IPCC 2021 [5] and the indicator is GWP100.

Boundary. Inside the boundary are the node the job holds for its walltime, and the share of the interconnect, the storage and the site that serve it. The last three enter through the power usage effectiveness (PUE) declared in Section III-B. Outside the boundary are the author's laptop, the login node, the network between laptop and cluster, the one-off download of the input, the development of the code, and every node and job the allocation does not reserve.

Life cycle phases. For the compute node, the study keeps production, including the extraction of raw materials that cradle-to-gate data contain, and use. For the interconnect,

the storage and the site, it keeps use only, through the declared PUE. End of life is not inventoried separately: the disposal route of Aion’s hardware is not published. One of the candidate hardware datasets of Section III-B nonetheless includes disposal. That mismatch is listed in Section VI.

Components. The study is capped at four components, which Section III-B inventories in this order: (1) the *compute node*: its electricity during the walltime and a share of its manufacture; (2) the *interconnect*: the HDR100 InfiniBand fabric that carries the job’s reads and writes between the node and shared storage; (3) the *storage*: the shared GPFS file system that holds the input, the code and the output; (4) the *site*: cooling and power distribution around the machine.

B. Life cycle inventory

The inventory lists the physical flows the assessment runs on, and how each was obtained. It holds no grams: those belong to Section III-C. Every entry is gathered in Table I in Appendix A, with its source and the date it was read. The raw scheduler records, logs and commands are kept in the project repository.

The measurement. The frozen commit and input of Section III-A were run five times as separate batch jobs on 29 September 2026, submitted consecutively with `--dependency=singleton` so that no two overlapped. Each run held its node exclusively, and all five landed on `aion-0127`. All five completed with exit code 0 and wrote identical output. All five are kept and the median is reported. Walltimes were 362, 364, 368, 369 and 372 s: a median of 368 s and a spread of 10 s from fastest to slowest. File system caches were neither reset nor measured. The first run is not the slowest (the third is), so these runs show no first-run effect, but they do not establish whether any run started cold. The records were read with `sacct` on 29 September 2026. The fields read, and why each is the quantity claimed, are: `Elapsed`, the time from start to end of the allocation, which is the walltime the node is held; `AllocNodes` and `AllocCPUS`, the node and cores held; `ReqMem`, the memory reserved; `MaxRSS` of the Python step, the peak resident memory the program touched, which agrees with the maximum resident set size that `/usr/bin/time -v` reports to within 0.2%; `MaxDiskRead` and `MaxDiskWrite` of the batch and Python steps; and `ConsumedEnergy` and `ConsumedEnergyRaw`.

Whether an energy figure comes back. It does not come back from the scheduler. On 29 September 2026, `scontrol show config` reported the energy plugin `acct_gather_energy/rapl`. Nevertheless, both energy fields were empty for every step of all five jobs, and for a separate interactive allocation checked the same day. The empty fields are missing data, not zero energy. The processor’s own RAPL counters are readable by an ordinary user through `/sys/class/powercap`, while `perf` reports its package energy event as not supported. The job was therefore run five more times on 29 September 2026, with the same program, input and allocation. Each run read both package counters

directly before and after the Python step, and corrected for counter wrap. `aion-0127` was busy, so these five runs and an idle reading were moved to `aion-0144`, an identical regular node. All five completed with exit code 0 and wrote the same histogram, and Slurm’s energy fields stayed empty. Over Python steps of 357.6 to 366.4 s, the two CPU packages drew a median of 67 834 J (range 67 129 to 68 827 J), an average of 188 W. Read the same way for 60 s after the fifth run, the idle node drew 10 959 J, or 183 W. Assuming that idle rate held throughout each run, the job’s activity adds a median of only 1 863 J (range 1 690 to 1 953 J). So 97 % of the package energy is drawn by a node the job holds but barely uses. RAPL covers the processor packages only: memory, board, local disk, fans and network card are outside it, and so is the time the allocation runs outside the Python step. Because no instrument reaches the whole node, the node’s electricity is also estimated with the Green Algorithms model [6]: the walltime times the power of the cores and memory held, using 280 W per 64-core processor [4] and 0.3725 W per GB of memory. The whole 256 GB installed is counted, because the exclusive allocation holds all of it. Counting one active core gives 36.7 kJ (0.0102 kWh) per run. Counting all 128 allocated cores at full power gives 241 kJ (0.0670 kWh). The measured package energy alone, 67.8 kJ, is almost twice the first figure. So the one-core model understates the node, and both model figures are scenarios, not bounds. The measured package energy is the figure this inventory carries forward.

Allocation rule. Anything shared across many runs is divided down to one run by the node-seconds the job holds over the node-seconds the hardware delivers during its service life. For one exclusive node this is the walltime over the service life. The same rule is applied in every subsection below.

1) *Compute node:* Operational: 67 834 J of CPU-package electricity per run (median, RAPL), of which 1 863 J is above idle, over a median walltime of 368 s, holding 1 node and 128 cores. Memory reserved: 224 000 MiB (`ReqMem`), of the 256 GB installed [4]. Memory touched: 7 760 436 KiB, or 7.40 GiB (median `MaxRSS`). The job therefore touches 3.4 % of what it reserves, and 211 GiB stays reserved but unused. Embodied: `ecoinvent cut-off v3.11` [7] holds no server production activity. Searches on 29 September 2026 for *server* (58 activities, 32 products), *computer production*, *computer, desktop* and *computer, laptop* found none. The closest dataset is *computer production, desktop, without screen* (GLO, 1998–2024, per unit). It models a 2002 single-processor desktop of 11.3 kg and includes packaging and disposal. Under IPCC 2021 GWP100 it carries 221 kg CO₂e per unit. A two-socket AMD EPYC server, the Dell PowerEdge R7525, has a published production footprint of 1 768.4 kg CO₂e, of which 854.7 kg comes from two 4 TB SSDs [8]. That study uses IPCC 2013 rather than IPCC 2021, and its server differs from an Aion blade. Neither figure describes an Aion node. Which one enters Section III-C, and why, is decided there. Aion entered production in October–November 2021 and still runs in September 2026 [4]. No retirement date is published, so a service life of six years is assumed (Section VI). One

run then holds 368 s of 189 345 600 node-seconds, a share of 1.94×10^{-6} of the node.

2) *Interconnect*: The job runs on one node, so no node-to-node traffic occurs. Its only traffic on the HDR100 fabric (link rate 100, read with `ibstat`) is file system I/O. The step counters do not measure storage traffic directly. The Python step's `MaxDiskRead` of 680.67 MiB equals the decompressed stream that `gzip` writes into the pipe (713 674 148 bytes, or 680.61 MiB). The batch step's `MaxDiskWrite` of 680.72 MiB is `gzip` writing that same pipe. What the batch step reads beyond the pipe, about 198 MiB, is the 179 MiB compressed input plus about 19 MiB of other files, most likely the loaded modules. The data read from storage per run is therefore about 187 693 955 bytes plus about 19 MiB, and the data written is the 1 972-byte histogram plus the job log. This breakdown is inferred from the counters, not measured on the fabric. Runs 2 to 5 may have read the input from the node's page cache. The fabric's energy is not metered per job. It enters through the declared PUE of the site subsection.

3) *Storage*: The repository, input and output sit on Aion's shared GPFS file system (`df -T`, 29 September 2026) [4]. Per run, the job keeps the 187 693 955-byte input and writes the 1 972-byte output, reading and writing the bytes listed under the interconnect. Storage energy is not metered per job. It enters through the declared PUE of the site subsection.

4) *Site*: The University of Luxembourg HPC centre publishes no PUE for Aion. This study therefore declares 1.36, the EU average PUE for 2023, reported in 2024 and published in the European Commission's first technical report on data-centre energy performance [9], read on 29 September 2026. It is a regional proxy, not a measurement of this site. It multiplies the operational energy of the compute node, and covers cooling, power distribution, interconnect and storage. How far a different value would move the result is discussed in Section VI.

C. Life cycle impact assessment

D. Interpretation

E. Reporting

IV. OPTIMISATION

A. Optimisation procedure

B. Implementation

C. Footprint after optimisation

D. What generalises

V. CARBON PAYBACK

A. What this means

VI. THREATS TO VALIDITY

AI USAGE TRANSPARENCY

REFERENCES

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[9] F. Heatubun, D. Dragan, M. Bolchi, and M. Lopes Bautista, "Assessment of the energy performance and sustainability of data centres in EU: First technical report," European Commission, Directorate-General for Energy; Publications Office of the European Union, Luxembourg, Tech. Rep., Jul. 2025, prepared by EY with Borderstep and AIT. Figure 6, p. 34: EU average PUE 1.36 for 2023 data. Accessed 29 September 2026.

APPENDIX A

FIGURES AND TABLES

TABLE I
LIFE CYCLE INVENTORY OF ONE RUN (SECTION III-B). MEASURED ROWS ARE THE MEDIAN OF FIVE RUNS ON AION-0127, EXCEPT THE RAPL ROWS (AION-0144); ALL ROWS READ ON 29 SEPTEMBER 2026 UNLESS STATED.

Component	Quantity	Value and unit	Source	Kind
Compute node	Walltime	368 s (range 362–372)	sacct Elapsed, jobs 15860349–53	measured
	Nodes, cores held	1 node, 128 cores	sacct AllocNodes, AllocCPUS	measured
	Memory reserved	224 000 MiB	sacct ReqMem	measured
	Memory touched	7 760 436 KiB (7.40 GiB)	sacct MaxRSS, step .0	measured
	Electricity, scheduler	none returned	ConsumedEnergy (Raw) empty	measured
	CPU-package energy per run	67 834 J (range 67 129–68 827)	RAPL, jobs 15861652–60	measured
	CPU-package power, idle	183 W (10 959 J in 60 s)	RAPL, job 15861674	measured
	Package energy above idle	1 863 J (range 1 690–1 953)	run — idle rate × step time	derived
	Electricity, 1 core active	36.7 kJ	Green Algorithms [6]	model
	Electricity, 128 cores at TDP	241 kJ	Green Algorithms [6]	model
	Hardware	2 × EPYC 7H12, 256 GB	[4], lscpu	looked up
	Manufacture, ecoinvent proxy	221 kg CO ₂ e per unit	[7], desktop without screen	looked up
	Manufacture, vendor proxy	1 768.4 kg CO ₂ e per unit	[8], R7525, IPCC 2013	looked up
	Service life	6 years	Section VI	assumption
	Share per run	1.94×10^{-6}	368 s / 189 345 600 s	derived
Interconnect	Bytes read from storage	≈ 187.7 MB + 19 MiB	MaxDiskRead, batch minus pipe	inferred
	Bytes written to storage	1 972 B + log	stat of output	measured
Storage	Input stored	187 693 955 B	stat, SHA-256 checked (28 Sep)	measured
	Output stored	1 972 B	stat	measured
Site	PUE	1.36	EU average, 2023 data [9]	declared proxy
All	Runs per year	100	Section VI	assumption

APPENDIX B
AI USAGE JOURNAL

AI model: Claude Opus 5.5 (claude-opus-5-5)

Interaction: chat, through the Claude desktop app

Prompts: 17

Input tokens: 13 418 158, mostly re-read context **Output tokens:** 88 944

Energy factor: 1.03 Wh per 1000 output tokens, so 91.6 Wh

Estimated: converted to gCO₂e with the grid factor declared in Section III-C

Kinds of task, and which sections they touched: I used the chat as a guide while I set up the repository and the job, and then for general corrections and fact checking of Sections III.A and III.B and of the references. All the text was spell-checked and proofread with it. The choices of scope, boundary and measurement were mine, taken step by step with the chat.

Wrong information:

- 1) it split the commit hash of the job wrongly when breaking it over lines. I compared it with the repository and corrected it.
- 2) it described the end-of-life treatment of the cut-off model wrongly. I rewrote the sentence.
- 3) some reference details were wrong: the release year ofecoinvent 3.11 and the data year of the EU PUE. I checked each reference at its source.